1. Suppose you are deciding upon investment options in stock markets. You have the following investment options:

Investment Portfolio	**Favorable Market**	**Unfavorable Market**
Company X	50,000	-60,000
Company Y	10,000	-4,000
50% Company X, 50% Company Y	30,000	-32,000

Also, you have the option of not investing at all. What will be the action taken by regret minimization agent?

a. Invest in Company X
b. Invest in Company Y
c. Invest 50% in Company X, Invest 50% in Company Y
d. Do not invest

Investment Portfolio	**Favorable Market**		**Unfavorable Market**		**Max Regret**
	Reward	**Regret**	**Reward**	**Regret**	
Company X	50,000	0	-60,000	60,000	60,000
Company Y	10,000	40,000	-4,000	4,000	40,000
50% Company X, 50% Company Y	30,000	20,000	-32,000	32,000	32,000
Don't invest	0	50,000	0	0	50,000

2. Suppose you are deciding upon investment options in stock markets. You have the following investment options:

Investment Portfolio	**Favorable Market**	**Unfavorable Market**
Company X	50,000	-60,000
Company Y	10,000	-4,000
50% Company X, 50% Company Y	30,000	-32,000

Also, you have the option of not investing at all. Suppose you know that markets are favorable with a probability 'p'. What is the minimum value of p for which a rational agent will invest? (Round off to 1 decimal point e.g. if you answer is 3.38 then write 3.4)

Ans. 0.3

Investment Portfolio	**Favorable Market**	**Unfavorable Market**	**Expected Reward**
Company X	50,000	-60,000	110,000p - 60,000
Company Y	10,000	-4,000	14,000p - 4,000
50% Company X, 50% Company Y	30,000	-32,000	62,000p - 32,000

For rational agent to invest,
max(110,000p - 60,000,14,000p - 4,000, 62,000p - 32,000) > 0

p > 60/110, p>4/14, p>32/62 or p>0.5454, p>0.2857, p>0.5161

min value of p for which a rational agent will invest 0.3.

3. Suppose you are deciding upon investment options in stock markets. You have the following investment options:

Investment Portfolio	**Favorable Market**	**Unfavorable Market**
Company X	50,000	-60,000
Company Y	10,000	-4,000
50% Company X, 50% Company Y	30,000	-32,000

Also, you have the option of not investing at all. The market can be favorable or unfavorable with equal probability. Let expected value of perfect information be v. Write v/1000.

Ans. 22

Investment Portfolio	**Favorable Market**	**Unfavorable Market**	**Expected Reward**
Company X	50,000	-60,000	-5,000
Company Y	10,000	-4,000	3,000
50% Company X, 50% Company Y	30,000	-32,000	-1,000
Do not invest	0	0	0

Expected reward with Perfect Information = (50,000+0)/2 = 25,000
Expected value of Perfect Information = 25,000 - 3,000 = 22,000

4. Why is discount factor used in an MDP formulation?
   a. **To ensure that cumulative rewards in infinite horizon MDP do not diverge**
   b. To speed up the computation
   c. **To capture the intuition that rewards in present are more important than rewards in future**

Factual question. Covered in class slides

5. How many deterministic policies are possible in an MDP with S states, where agent can take A different actions in each state?
    a. **A^S**
    b. S^A
    c. S.A
    d. $2^{S.A}$

A policy can be defined as a tuple of $(A_1, A_2, \ldots A_s)$ where A_i is the action policy suggests in state i. Each state has A possible actions. So number of possible tuples will be A^S.

6. Suppose you are deciding upon investment options in stock markets. You have the following investment options:

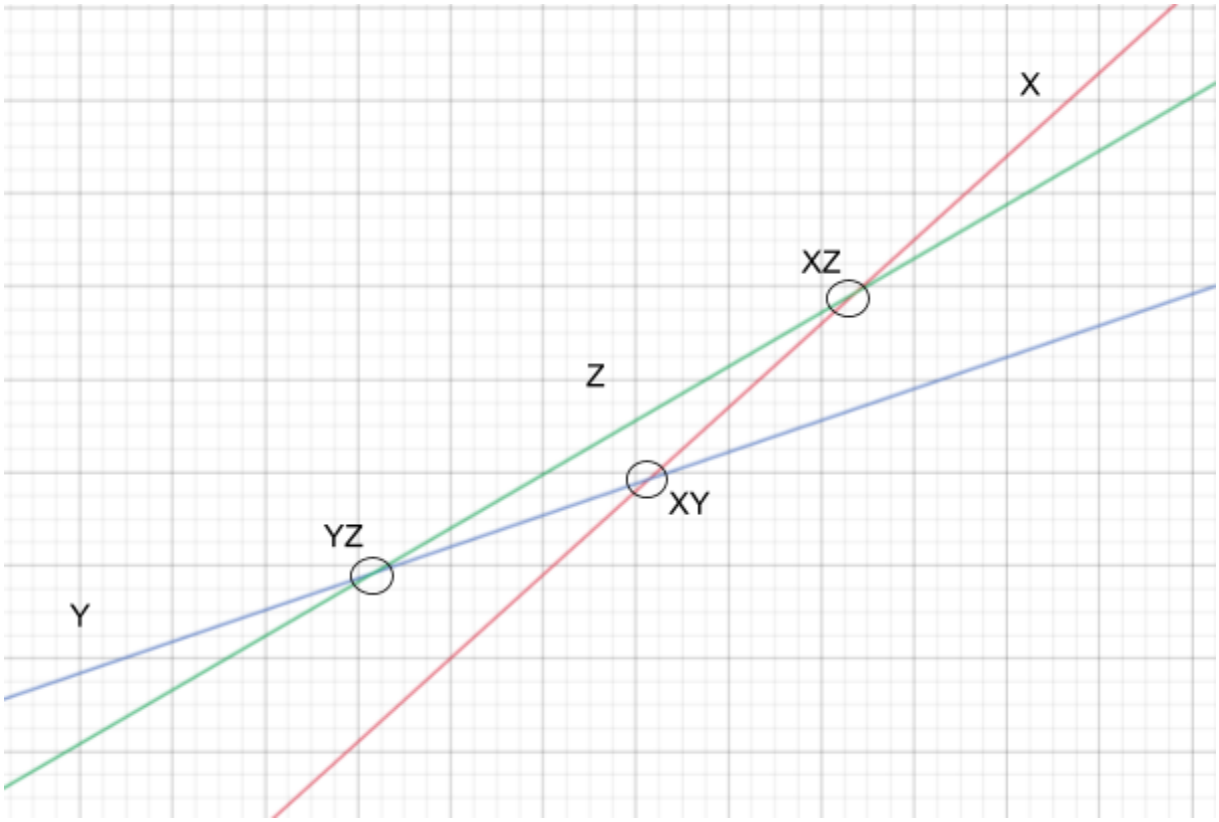
Investment Portfolio	**Favorable Market**	**Unfavorable Market**
Company X	40,000	-50,000
Company Y	20,000	-14,000
Company Z	30,000	-28,000

Using Hurwicz criterion for decision making using coefficient of realism (α), which of the following statement(s) is/are correct? (Hint: Visualize graphically). Assume that investing in some company is necessary.

    a. **At α >= 7/12, the agent changes its decision to invest in Company Y to Z.**
    b. At α >= 9/14, the agent changes its decision to invest in Company Y to X.
    c. **At α >= 11/16, the agent changes its decision to invest in Company Z to X.**
    d. **Independent of the stock prices of companies X, Y and Z, the agent changes its decision for at most two values of α.**

Consider the coefficient of realism α.
Expected reward for Company X = $40000\alpha - 50000(1 - \alpha) = 90000\alpha - 50000$
Expected reward for Company Y = $20000\alpha - 14000(1 - \alpha) = 34000\alpha - 14000$
Expected reward for Company Z = $30000\alpha - 28000(1 - \alpha) = 58000\alpha - 28000$

Consider the graphs of these equations:
1. y = 90000x - 50000 (Red)
2. y = 34000x - 14000 (Blue)
3. y = 58000x - 28000 (Green)



Point YZ = (7/12, 70000/12)
Point XY = (9/14, 110000/14)
Point XZ = (11/16, 190000/16)

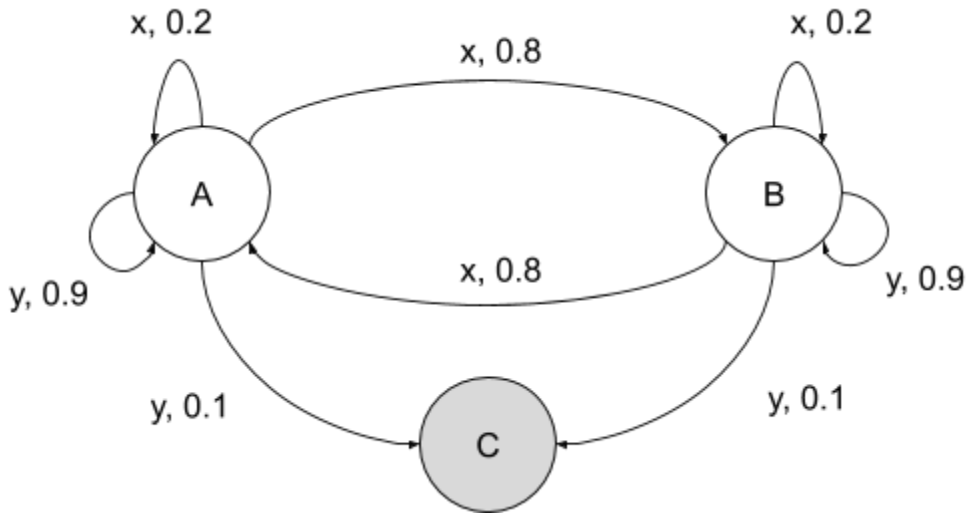
We consider the function
$f(\alpha) = \max(90000\alpha - 50000, 34000\alpha - 14000, 58000\alpha - 28000)$
The function f shifts from blue to green i.e. Y to Z at $\alpha = 7/12$.
The function f shifts from green to red i.e Z to X at $\alpha = 11/16$.
Note that there is no shift from blue to red at Point XY since the max is attained by the green line.

Therefore A and C are correct and B is correct. It can be visualized intuitively that the number of such decision shifts can be at most two for any three lines on the coordinate plane. Therefore, D is correct.

7. If the utility of money m for an agent is e^m, then that agent is a
   a. **Risk-prone agent**
   b. Risk-averse agent
   c. Risk-neutral agent

Seeing the graph of $y = e^x$, it can be seen that the utility of positive money is much higher compared to the utility of negative money. Therefore, the agent is a risk-prone agent.

8. Consider an undiscounted MDP having three states A, B, C. Any transition that ends up in states A, B or C yields a reward of -1, -3, 0 respectively. C is the terminal state. Actions x, y are possible in the states A and B with transition probabilities shown in the diagram.



For the policy π of performing action y in both the states, find the value of the state B i.e $V^{\pi}(B)$?

Ans: -27

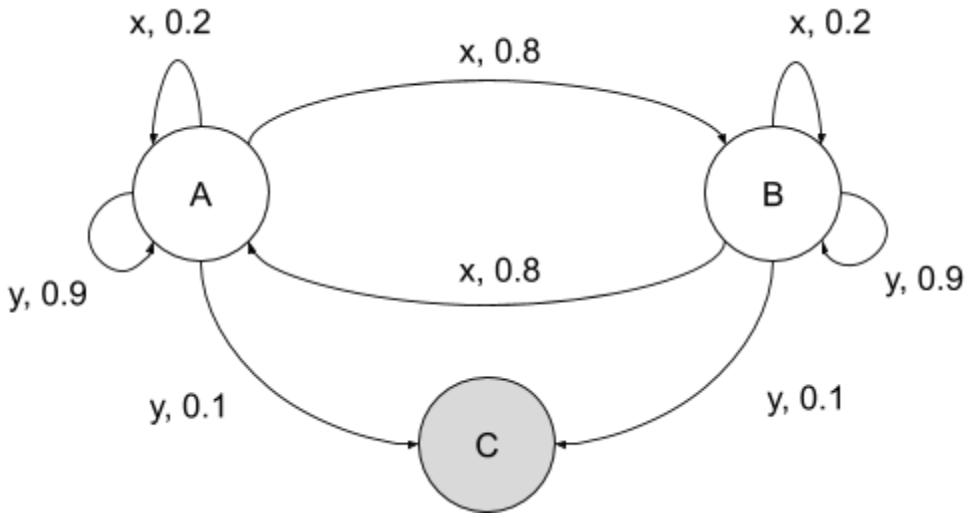
$V^{\pi}(B) = 0.9 * (-3 + V^{\pi}(B)) + 0.1 * (0 + V^{\pi}(C))$

Note that $V^{\pi}(C) = 0$ since C is a terminal state.

Therefore, we have
$V^{\pi}(B) = 0.9 * (-3 + V^{\pi}(B)) + 0.1 * (0 + 0)$
$V^{\pi}(B) = -2.7 + 0.9V^{\pi}(B)$
$0.1V^{\pi}(B) = -2.7$
$V^{\pi}(B) = -27$

9. Consider an undiscounted MDP having three states A, B, C. Any transition that ends up in states A, B or C yields a reward of -1, -3, 0 respectively. C is the terminal state.

Actions x, y are possible in the states A and B with transition probabilities shown in the diagram.



Starting with the initial policy π_0 of performing action y in both the states, perform one iteration of the policy iteration algorithm to obtain a policy π_1. Which of the following is true about the policy π_1?

a. Action x is favourable in both states A and B.
b. Action y is favourable in both states A and B.
c. Action x is favourable in state A and action y is favourable in state B.
d. Action y is favourable in state A and action x is favourable in state B.

Policy $\pi = \pi_0$.

$V^{\pi}(A) = 0.9 * (-1 + V^{\pi}(A)) + 0.1 * (0 + V^{\pi}(C))$
$V^{\pi}(B) = 0.9 * (-3 + V^{\pi}(B)) + 0.1 * (0 + V^{\pi}(C))$

Using $V^{\pi}(C) = 0$ and solving the above equations, we get
$V^{\pi}(A) = -9$, $V^{\pi}(B) = -27$

$Q^{\pi}(A, x) = 0.8 * (-3 + V^{\pi}(B)) + 0.2 * (-1 + V^{\pi}(A)) = 0.8 * (-30) + 0.2 * (-10) = -26.$
$Q^{\pi}(A, y) = 0.9 * (-1 + V^{\pi}(A)) + 0.1 * (0 + V^{\pi}(C)) = 0.9 * (-10) = -9.$

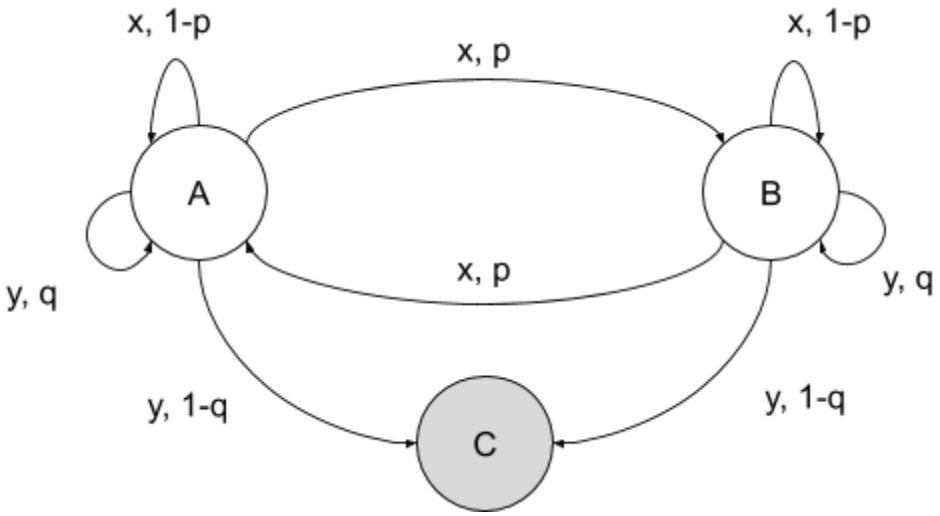
$Q^{\pi}(B, x) = 0.8 * (-1 + V^{\pi}(A)) + 0.2 * (-3 + V^{\pi}(B)) = 0.8 * (-10) + 0.2 * (-30) = -14.$
$Q^{\pi}(B, y) = 0.9 * (-3 + V^{\pi}(B)) + 0.1 * (0 + V^{\pi}(C)) = 0.9 * (-30) = -27.$

π_1 is the greed policy with respect to Q^{π}.
Therefore,

$\pi_1(A) = y$ since $Q^{\pi}(A, y) > Q^{\pi}(A, x)$.
$\pi_1(B) = x$ since $Q^{\pi}(B, x) > Q^{\pi}(B, y)$.

Therefore, D is correct.

10. Consider an undiscounted MDP having three states A, B, C. Any transition that ends up in states A, B or C yields a reward of -1, -3, 0 respectively. C is the terminal state. Actions x, y are possible in the states A and B with transition probabilities shown in the diagram. (Hint: Goal is to maximize the reward)



Which of the following statements is true for the optimal policy π^*?

a. Optimal action in state A is always x, irrespective of p and q.
b. Optimal action in state A is always y, irrespective of p and q.
c. None of the above

Since the reward in both the states A and B is negative and zero in C, the goal is to reach C as soon as possible to maximize the reward. Now, C can be reached from state A and B with an equal probability of q. Therefore the optimal action in state A should always be y since using the action x to go to B would never be favourable as the reward in B is less than that in the state A. Note that optimal action in state B is not independent of p and q. It might be favourable to go to A using action x and then transit to C (eg. when (1-q) is very low as in the previous question) or it might directly use action y to transit to C (eg. when (1-q) is very high).